Visor

USER CASES AND HANDS-ON TEST REPORT

How Visor holds up for students, advisors and the registrar

Twenty-five user cases, each run by hand against the working app: what the person wants, what we did, what Visor showed, and whether that was right. Findings at the end are ranked so the team can pick what to build next.

25	17	7	1
user cases	pass	pass with a note	fail

Build tested	**`nathanaelhub/visor`** at commit **`96d1672`** (31 Oct 2026)
Catalog	Lipscomb University 2026-27, B.S. Computer Science (127 hours)
Data	Five fictional demo students and illustrative equivalencies. No real student records.
Tested by	Nathanael Johnson (graduate assistant), with Claude
Builds on	Lunatic Labs Visor: Austin Patton, Gabriel Pop, Firas Adas, with Dr. Steve Nordstrom

How we tested

Unit tests (117, all passing) prove the rules engine works piece by piece. These cases test it the way a person would: start from someone's goal, click through the real screens, and check every number on screen against the catalog data and the API's own answer.

Angle	How it was run
Student, advisor, registrar	Local app (React on Vite, Flask API, SQLite seeded from `data/`) driven in a real browser, desktop at 1280 px wide. A screenshot was taken for each case.
Ask Visor	Live Claude Opus 5.5 through Visor's relay, using a server-side key so no key ever appears on screen. Every fact in each answer was checked against the catalog files.
Planner edge cases	Direct calls to `POST /api/audit`. Every plan returned includes the independent checker's result (`rules/verify.py`), which shares no code with the planner.
General	Landing page, dark mode, and a 390 px phone viewport.

Verdicts: **Pass** worked as intended. **Note** worked, but we saw something worth improving. **Fail** did the wrong thing. Finding numbers (F1…) point to the last section.

Summary

Case	Who wants what	Verdict	Findings
S1	New freshman sees every semester to graduation	Pass	
S2	Junior checks progress and graduation term	Note	F10
S3	Student understands why a course is back in the plan (retake)	Pass	
S4	Almost-done student sees the last term	Note	F9
S5	Student can see the plan but not change it	Pass	
A1	Advisor scans advisees and finds who needs attention	Pass	
A2	Advisor sees why a plan needs review	Note	F3, F12
A3	Advisor sees how a transfer student's credit counted	Note	F2
R1	Registrar looks up a confirmed equivalency	Pass	
R2	Registrar looks up a proposed equivalency	Pass	F13
R3	Registrar flags an unknown course for a professor	Note	F4
R4	Registrar mistypes a course code	Fail	F7
C1	Student connects an API key and picks a model	Pass	
C2	Student asks what to register for and why	Note	F5
C3	Student asks Ask Visor to move a course	Pass	
C4	Student pastes a bad key	Pass	
C5	Student asks whether transfer credit counts	Pass	
E1	Same record twice gives the same plan	Pass	
E2	Summer terms turned on	Note	F6
E3	Lower credit cap stretches the plan safely	Pass	
E4	Missing prerequisite gets added on its own	Pass	
E5	Bad input to the API	Pass	F8
G1	Visitor understands Visor from the landing page	Pass	F1
G2	Dark mode	Pass	
G3	Phone layout	Pass	F11

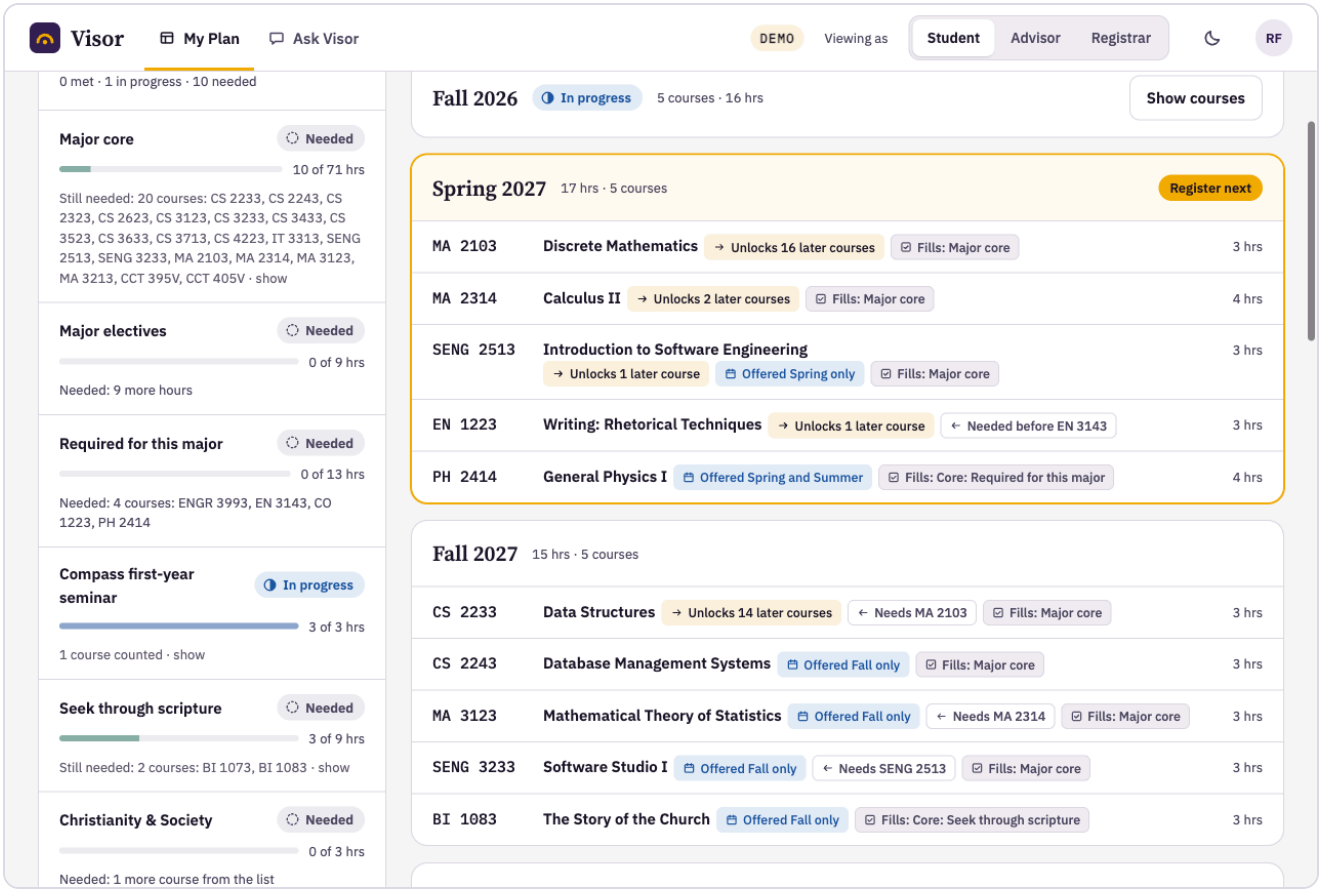
Student cases

Students see their own plan, read-only. Each demo student covers a different situation.

UC-S1 New freshman sees every semester to graduation

Pass

Who	Riley Fresh (fictional), first semester, five courses in progress, nothing finished.
Goal	“Show me what I take each semester until I graduate.”
Steps	1. Landing page → “I’m a student”. 2. Pick Riley Fresh in the avatar menu. 3. Read the summary and the semester plan.
Expected	Every remaining requirement placed or warned; graduation term shown; hours add up to 127.
Actual	16 hours in progress, then 7 planned terms (Spring 2027 to Spring 2030, 15–17 hours each), graduation **May 2030**. Plan checks all pass, no warnings. 16 + 107 planned + 4 free-elective hours = 127. Every course carries a reason chip (“Unlocks 16 later courses”, “Offered Spring only”, “Needed before EN 3143”).

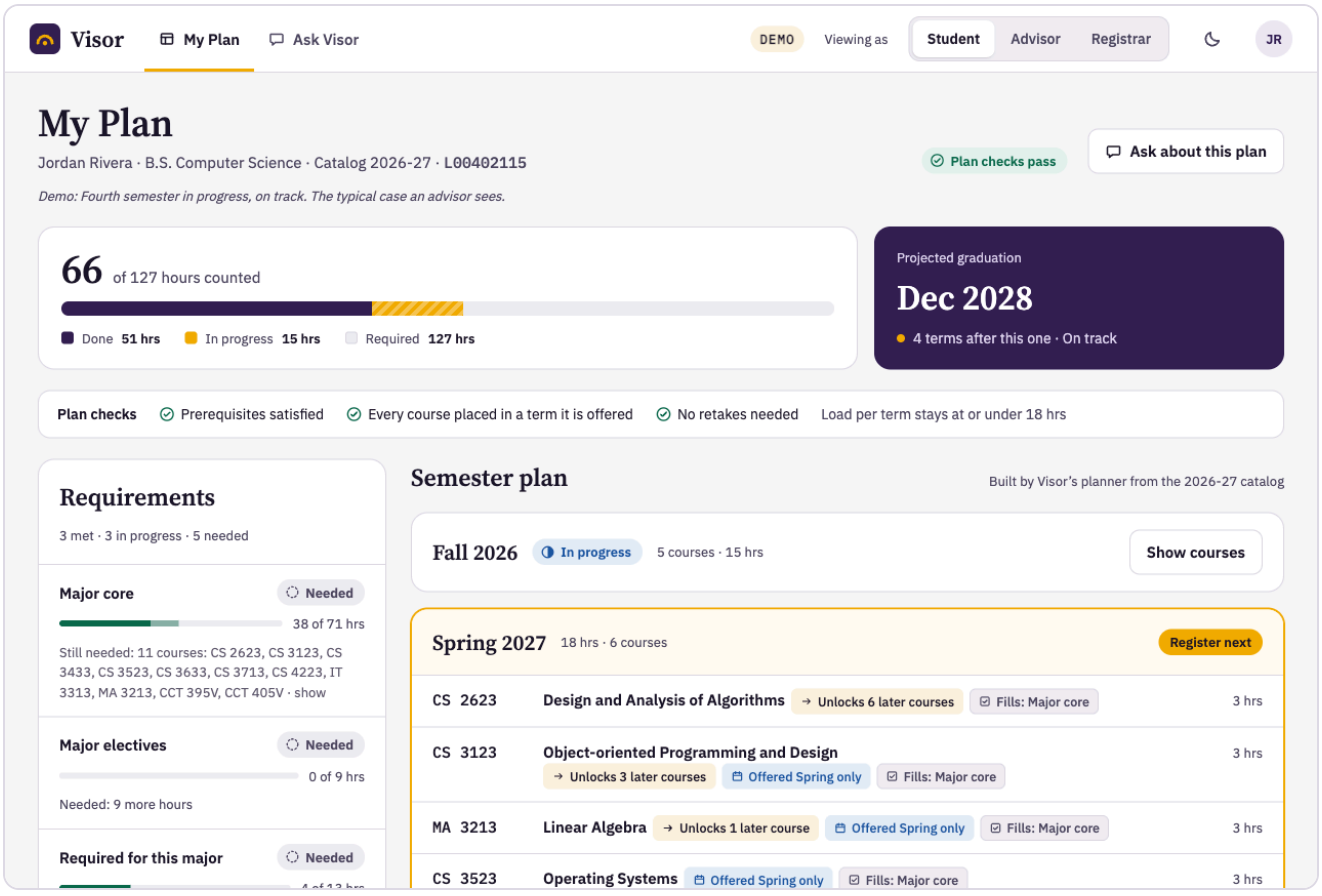


Riley’s plan: Spring 2027 is highlighted as “Register next”; every course says why it is there.

UC-S2 Junior checks progress and graduation term

Note

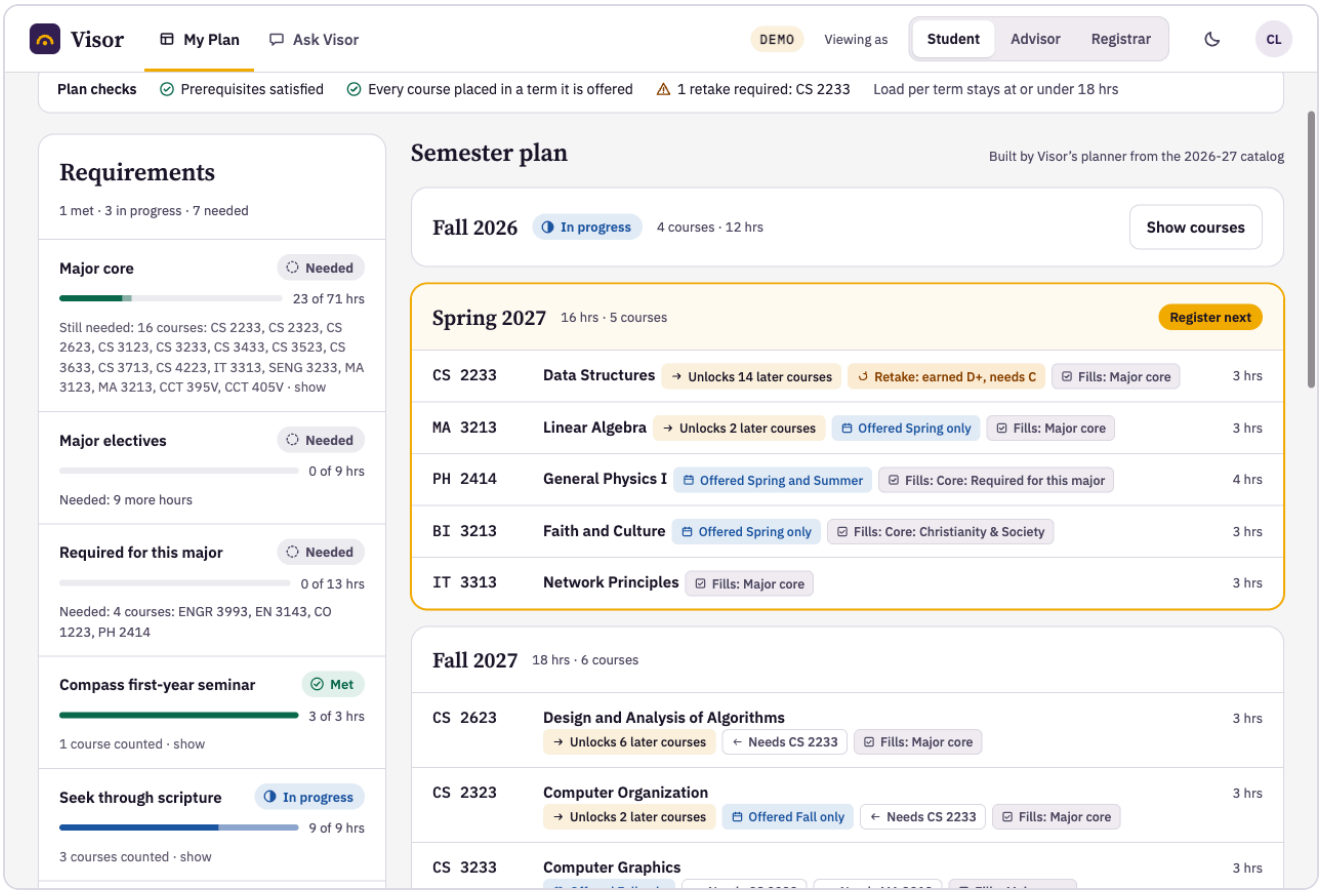
Who	Jordan Rivera (fictional), fourth semester in progress, on track.
Goal	“How far along am I, and when do I finish?”
Steps	1. Student view, Jordan Rivera (the default). 2. Read the progress card, plan checks and requirement list.
Expected	Done, in progress and remaining hours that match the audit; a graduation term.
Actual	66 of 127 hours counted (51 done, 15 in progress), graduation **Dec 2028**, “4 terms after this one · On track”. Requirements: 3 met · 3 in progress · 5 needed. Numbers match `/api/audit` exactly. **Note (F10):** “Major core” shows *Needed* at 38 of 71 hours. To a student that reads as “not started”.



UC-S3 Student understands why a course is back in the plan

Pass

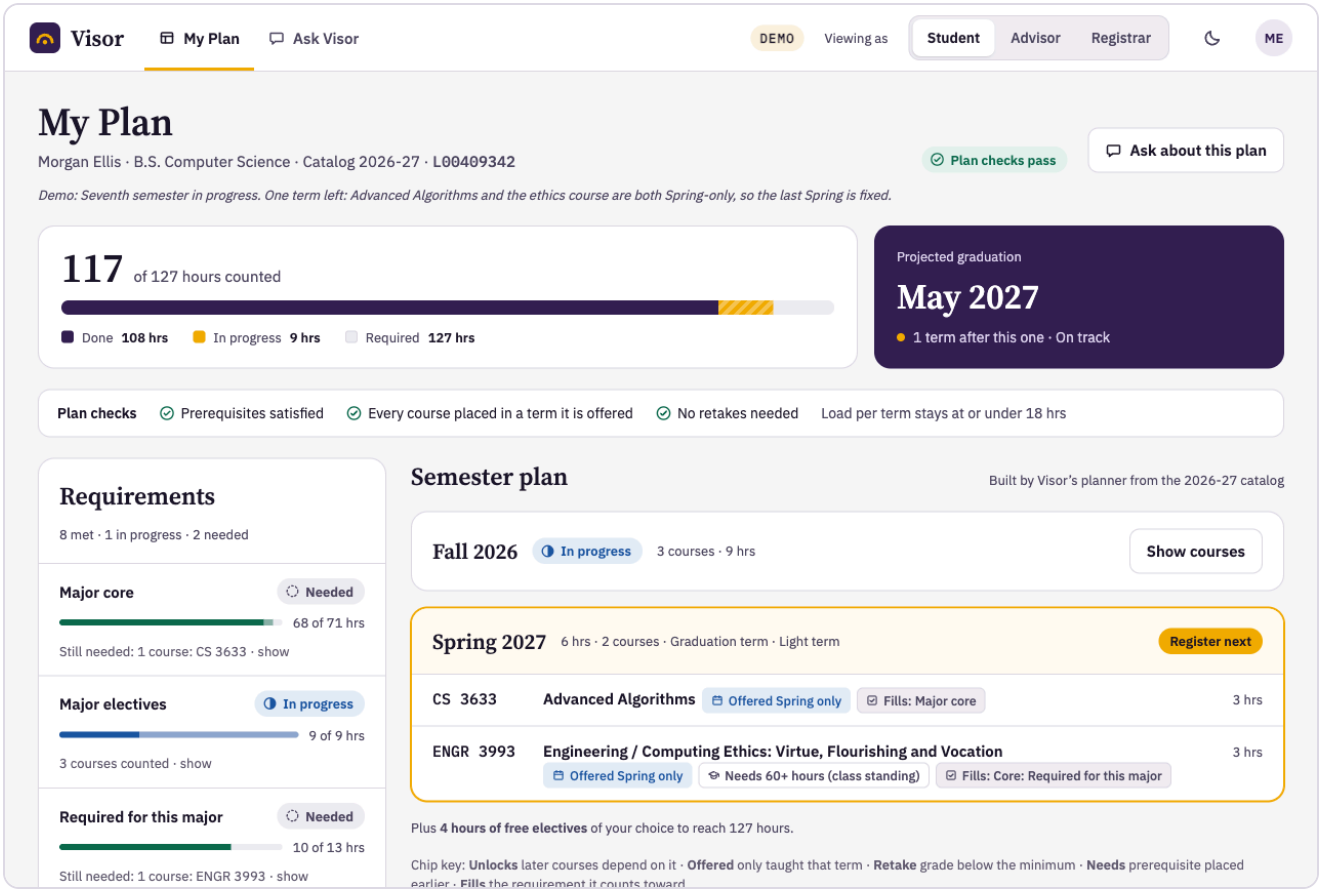
Who	Casey Lin (fictional): passed Data Structures (CS 2233) with a D+, but the major needs a C.
Goal	“Why is Data Structures in my plan again?”
Steps	1. Pick Casey Lin. 2. Read Plan checks and the Spring 2027 term.
Expected	The retake is placed first and says why; courses that depend on it wait.
Actual	Plan checks: “1 retake required: CS 2233”. CS 2233 sits first in Spring 2027 with “Retake: earned D+, needs C” and “Unlocks 14 later courses”. Graduation May 2029.



UC-S4 Almost-done student sees the last term

Note

Who	Morgan Ellis (fictional), 108 hours done, 9 in progress.
Goal	“What’s left, and is next semester my last?”
Actual	117 of 127 counted; Spring 2027 is the **graduation term** with CS 3633 and ENGR 3993 (6 hours, marked “Light term”); graduation May 2027. Both courses explain themselves (“Offered Spring only”, “Needs 60+ hours”). **Note (F9):** the page also says “Plus 4 hours of free electives”. Those 4 hours have to go somewhere, and the light last term is the obvious place, but the plan doesn’t suggest it. The “nothing left” case (plan empty, “complete pending final grades”) is covered by the unit test `test_nothing_left`; no demo student reaches it.



UC-S5 Student can see the plan but not change it

Pass

Goal	Students get a forecast, not an editor (the original requirement).
Steps	Listed every button, input and dropdown inside the student’s My Plan page.
Actual	Only two controls: “Ask about this plan” and “Show courses” (expands the in-progress term). There is nothing that edits the plan. Ask Visor also refuses edits (UC-C3).

Advisor cases

Advisors see every advisee, can open any plan, and should notice who needs attention first.

UC-A1 Advisor scans advisees and finds who needs attention

Pass

Steps

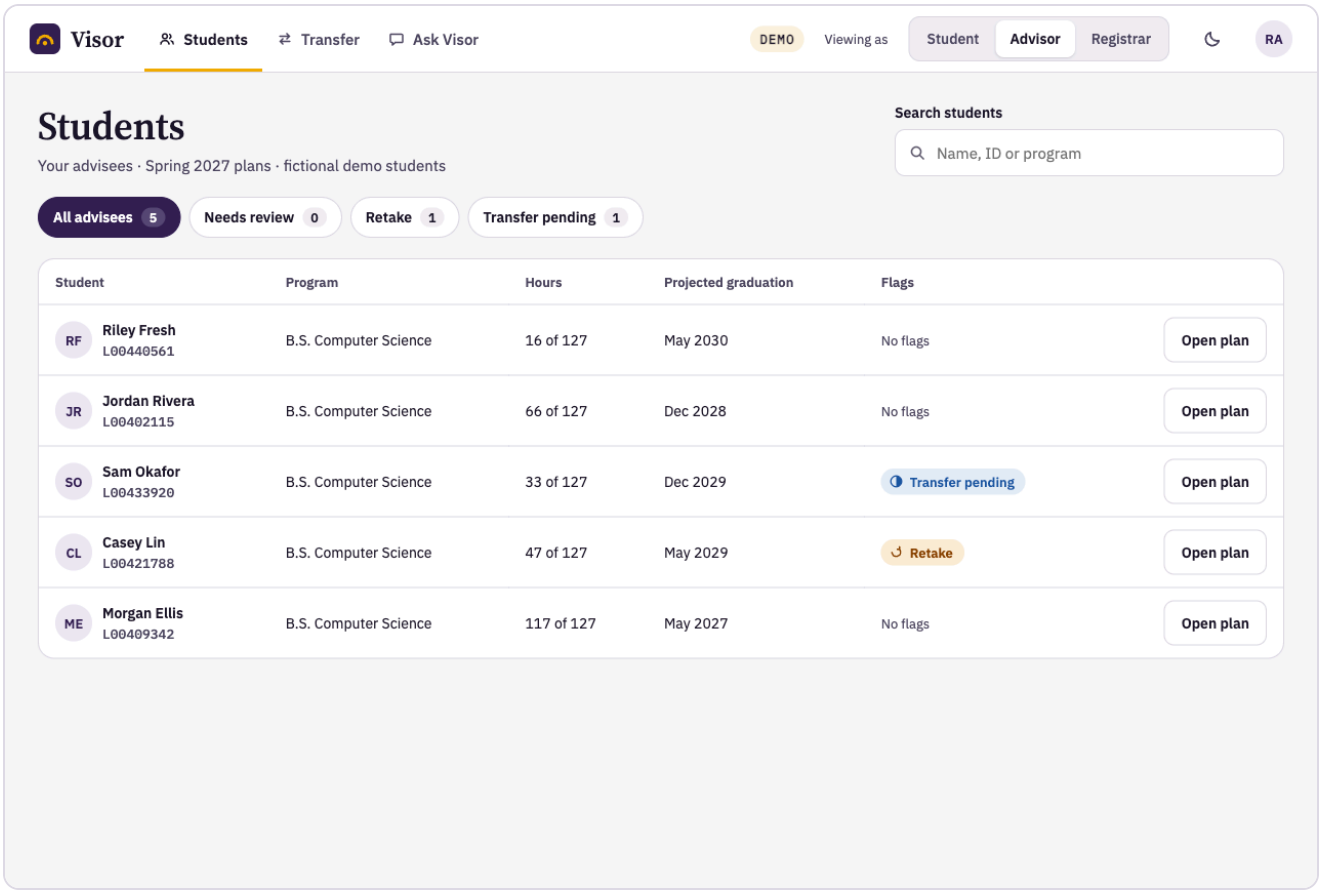
1. Switch “Viewing as” to Advisor → Students tab.
2. Use the filter chips and the search box.

Expected

All advisees with hours, graduation term and flags; filters and search narrow the list.

Actual

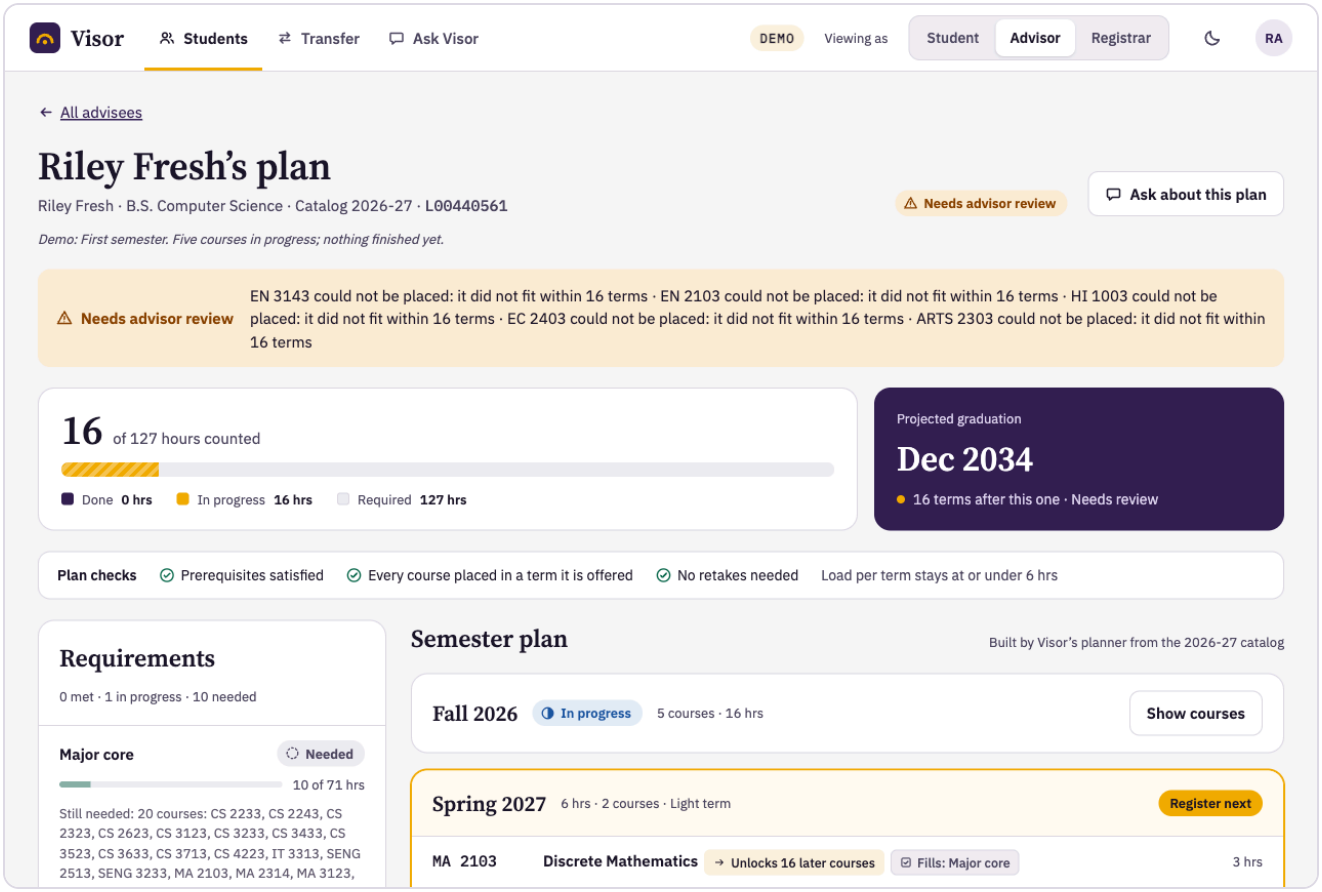
5 advisees with student numbers, hours and graduation terms that match each student’s My Plan. Chips: All 5 · Needs review 0 · Retake 1 (Casey) · Transfer pending 1 (Sam). “Transfer pending” shows only Sam; searching “casey” or the student number “L0043” finds the right person.



UC-A2 Advisor sees why a plan needs review

Note

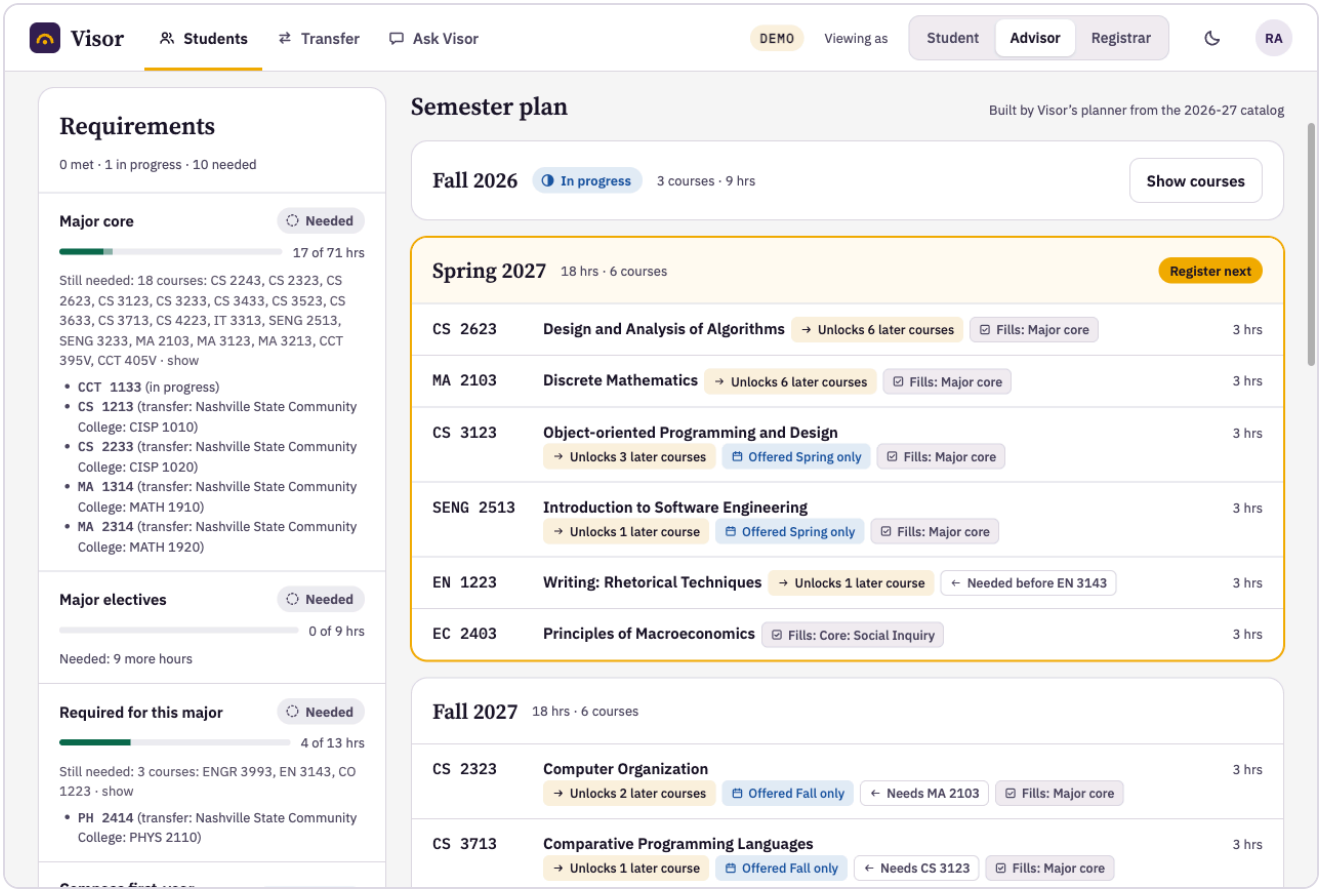
Setup	None of the five demo students produces a warning (F12), so we forced one by capping Riley's plan at 6 credit hours a term in the request the page sends.
Expected	"Needs advisor review" badge, each warning naming the course and what failed.
Actual	Badge and banner appear: "EN 3143 could not be placed: it did not fit within 16 terms", and four more like it. **Note (F3):** the card still says "Projected graduation **Dec 2034**", even though five required courses have no term. That date is not a real graduation date and should not be shown as one.



UC-A3 Advisor sees how a transfer student's credit counted

Note

Who	Sam Okafor (fictional): 10 transfer courses from Nashville State and Vol State, covering every status.
Steps	Students → filter "Transfer pending" → Open plan → expand requirement groups.
Actual	"33 of 127 · Includes 24 transfer hours." Counted courses show their source, e.g. "CS 2233 (transfer: Nashville State Community College: CISP 1020)". Elective-only credit adds hours, and free electives drop to 0. **Note (F2):** the three courses that *don't* count (CIS 2350 proposed, PHED 1110 not transferring, HIST 2010 unknown) appear nowhere on the page. The API returns them, and the advisee list flags "Transfer pending", but the advisor can't see *which* course is pending.



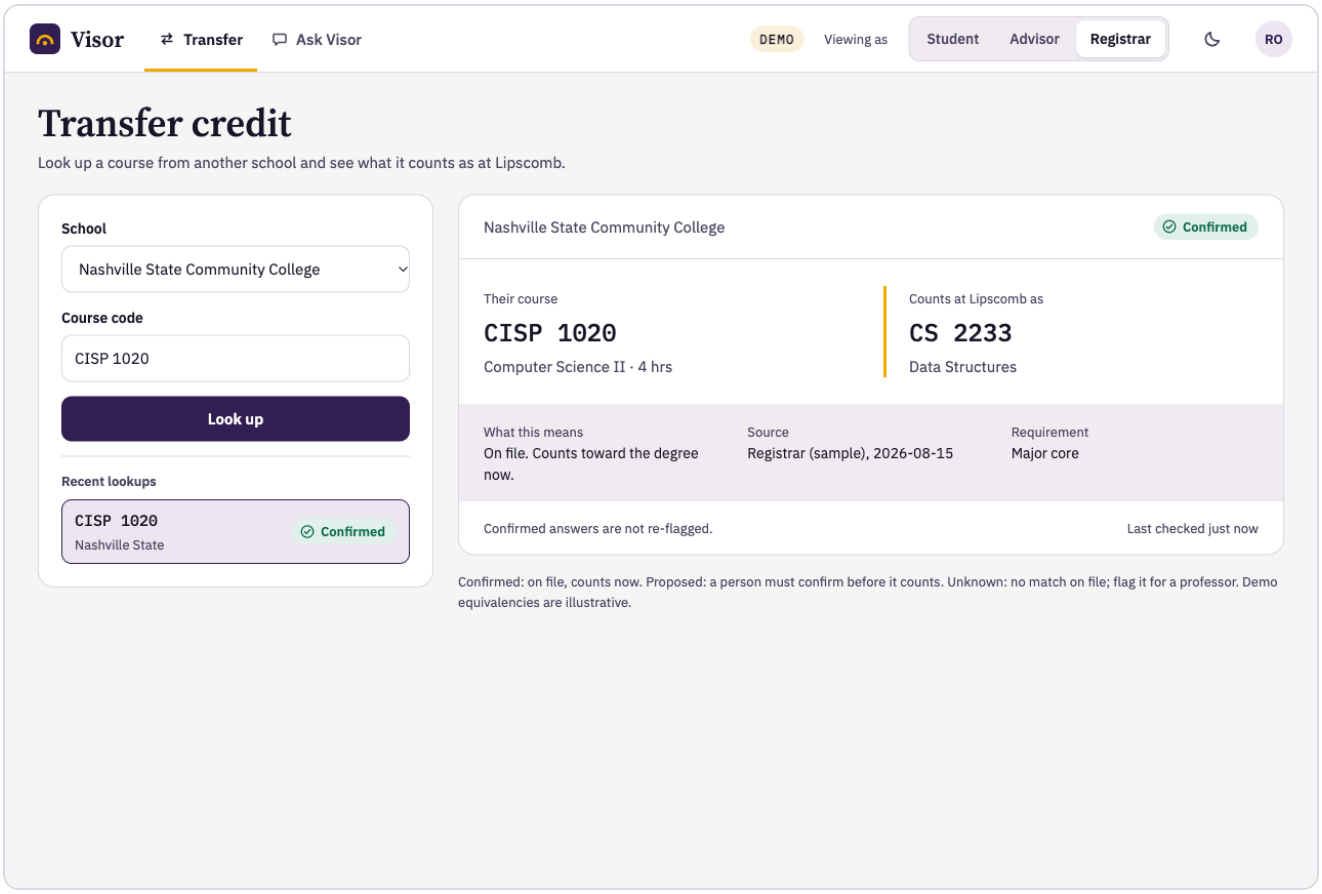
Registrar cases

The Transfer tab is the team's original feature (specs/03), shown with the new design. Lookups use the team's equivalency API unchanged.

UC-R1 Look up a confirmed equivalency

Pass

Steps	Registrar → Transfer → Nashville State, `CISP 1020` → Look up.
Actual	**Confirmed**: counts as CS 2233 Data Structures, Major core. Source "Registrar (sample), 2026-08-15". No flag button ("Confirmed answers are not re-flagged").

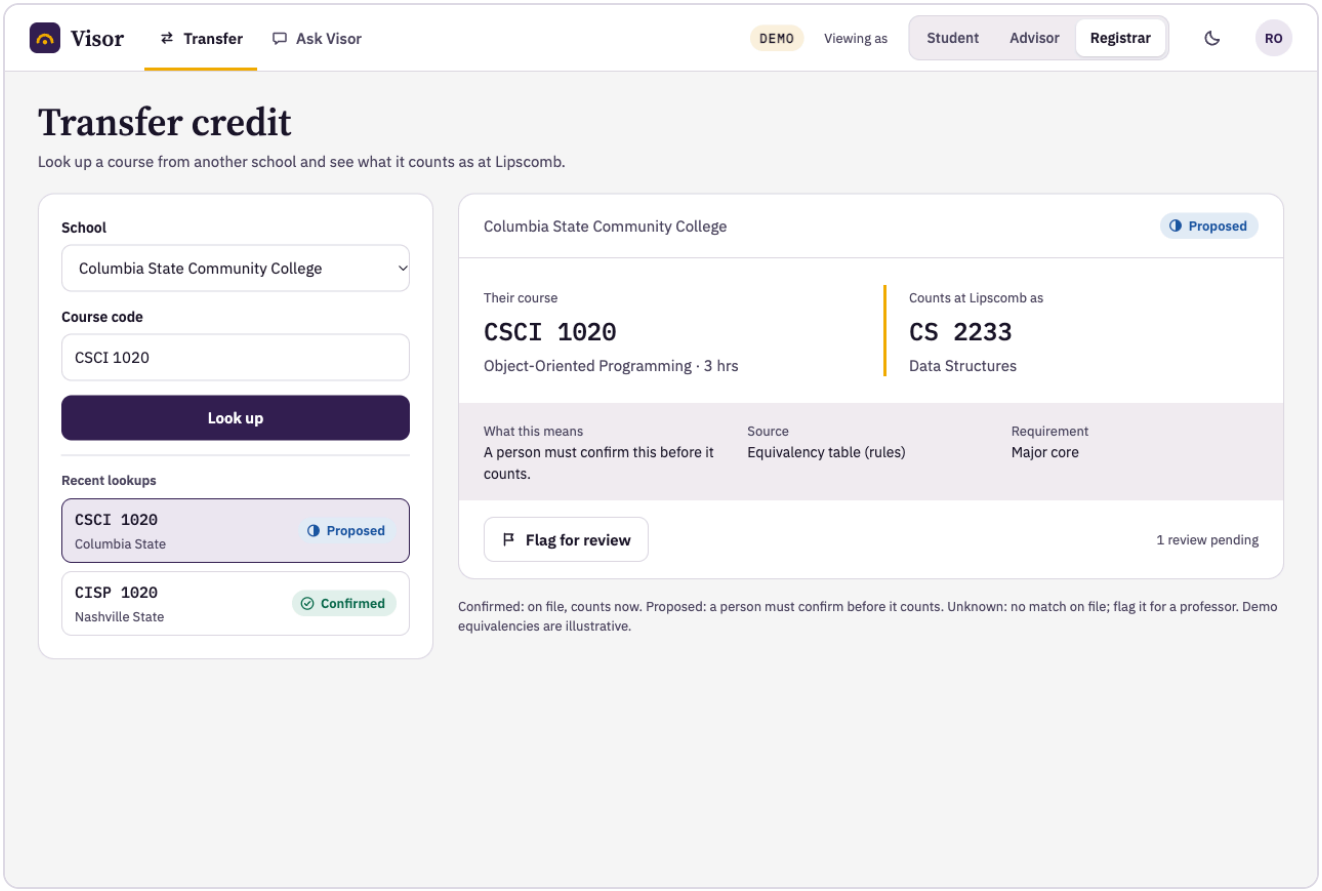


UC-R2 Look up a proposed equivalency

Pass

Steps Columbia State, `CSCI 1020`.

Actual **Proposed** → CS 2233: “A person must confirm this before it counts.” Source “Equivalency table (rules)”, “1 review pending”.
Data note (F13): the demo row maps *Object-Oriented Programming* to *Data Structures*. Ask Visor itself pointed out the mismatch (UC-C5).

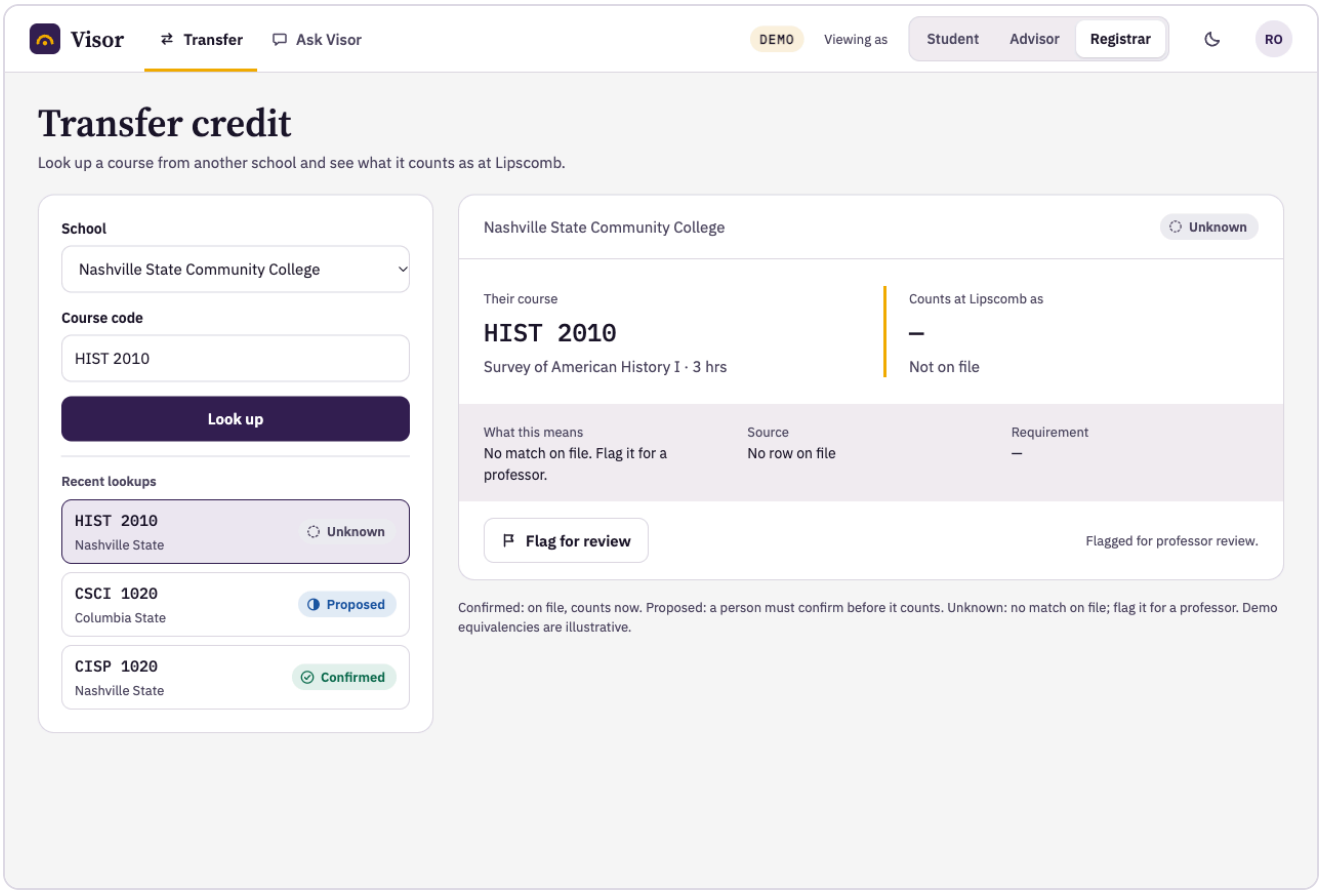


UC-R3 Flag an unknown course for a professor

Note

Steps Nashville State, `HIST 2010` → **Unknown** → Flag for review. Then click Flag again.

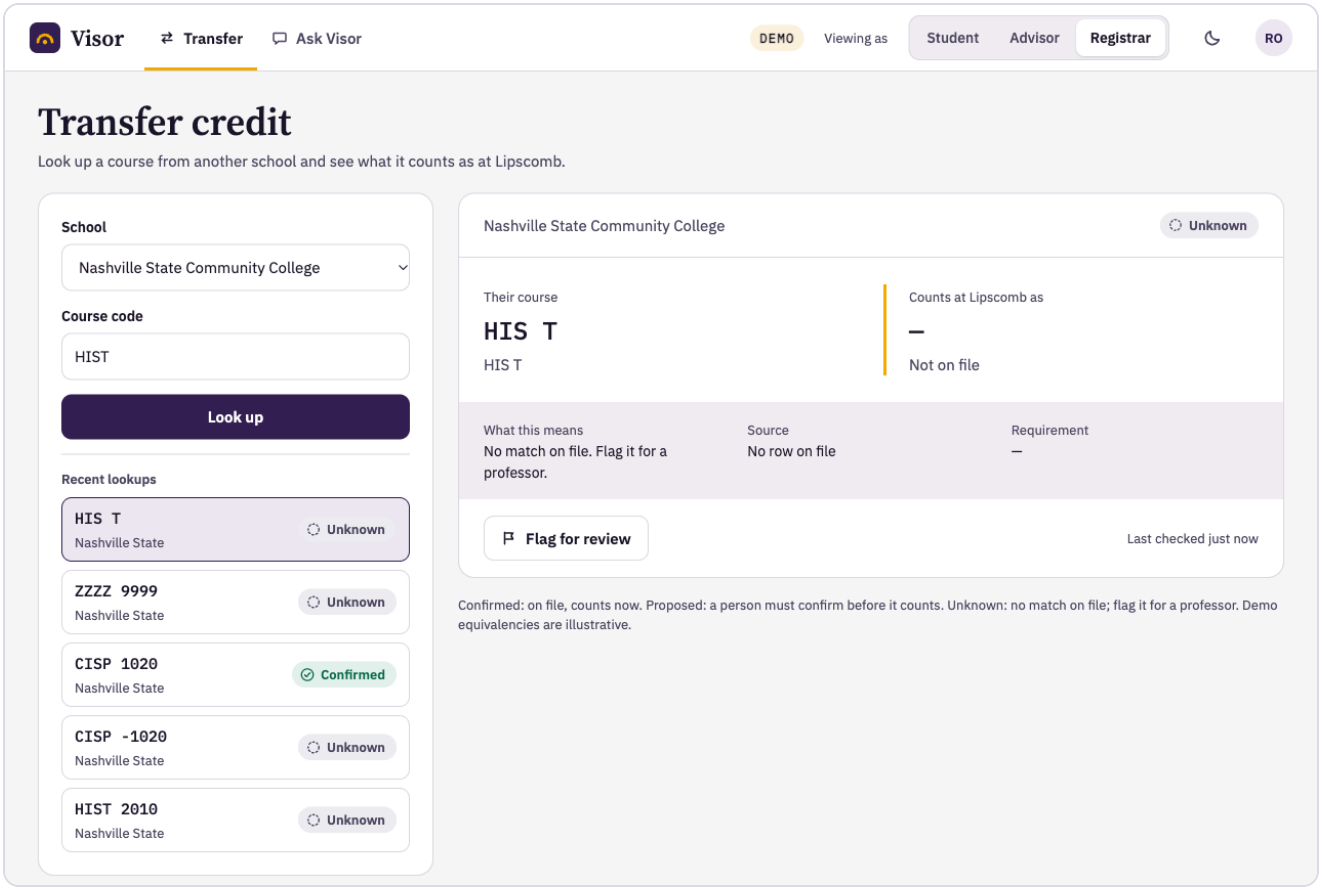
Actual First click: “Flagged for professor review.” The second click also succeeds, and the lookup now says “**2** review pending” (F4). After flagging, the Source line reads “Equivalency table (registrar)” for a course with no match, which is misleading.



UC-R4 Registrar mistypes a course code

Fail

Steps	Tried `HIST`, `12345`, `CISP-1020`, `cisp 1020`, `␣␣CISP␣␣␣1020␣`.
Expected	Clean-up of obvious variants; a clear message for anything else.
Actual	Lowercase and extra spaces work. But `HIST` is read as subject “HIS”, number “T”, and returns **Unknown: HIS T**, which a registrar could flag by mistake. `CISP-1020` becomes “CISP -1020” (Unknown, though CISP 1020 is confirmed). `12345` shows the right message, but the previous result stays on screen under it (F7).



Ask Visor cases

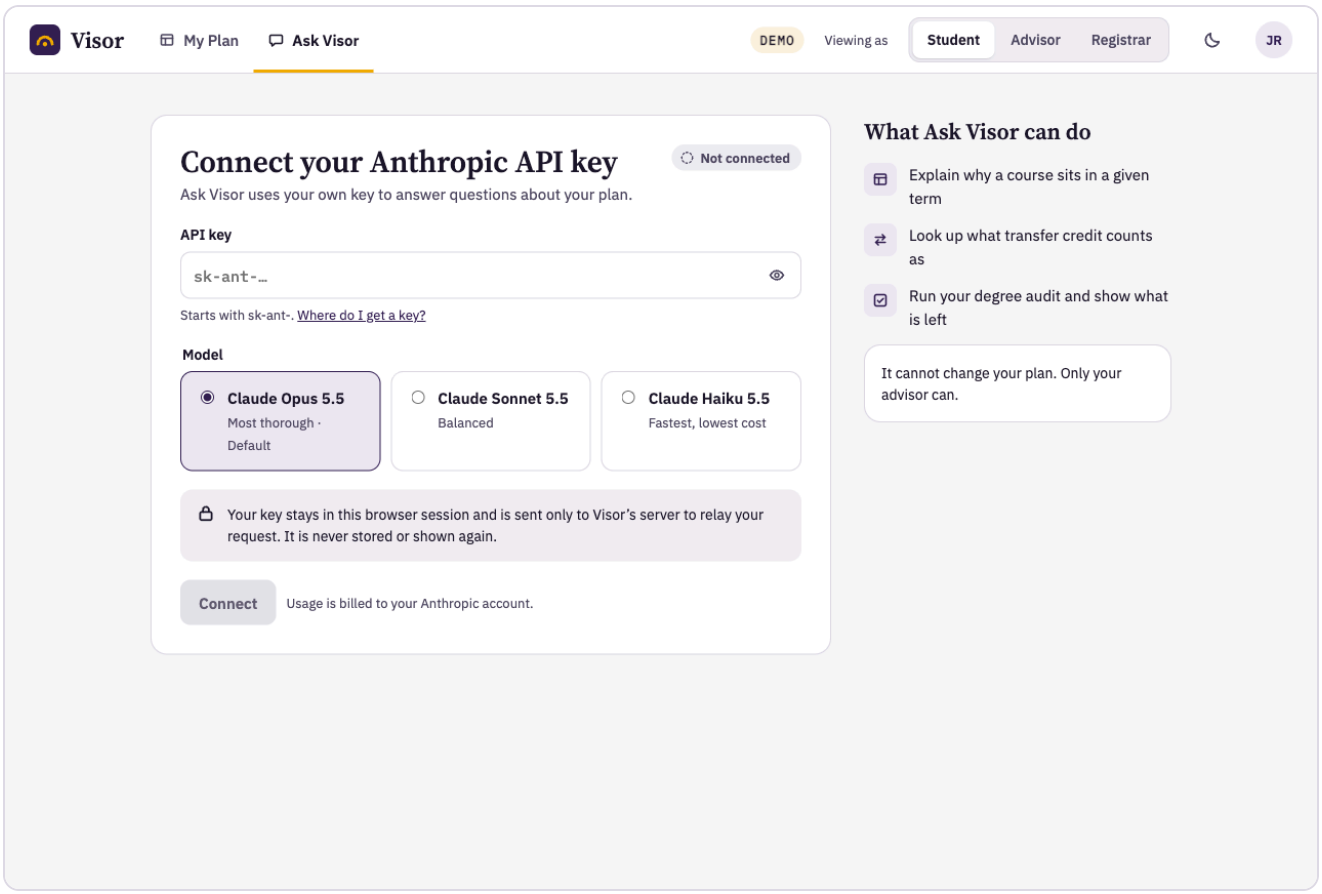
Claude answers through three read-only tools that call Visor's own code. It explains the plan and never changes it. Three live conversations as Casey Lin on Claude Opus 5.5. Each one answered in a few seconds, and we checked every factual claim against the catalog files.

UC-C1 Connect an API key and pick a model

Pass

Actual

Connect card with a masked key field, "Starts with sk-ant-" hint and link, three models (Opus 5.5 default, Sonnet 5.5, Haiku 5.5), and the promise "never stored or shown again". The right panel lists what Ask Visor can and cannot do.



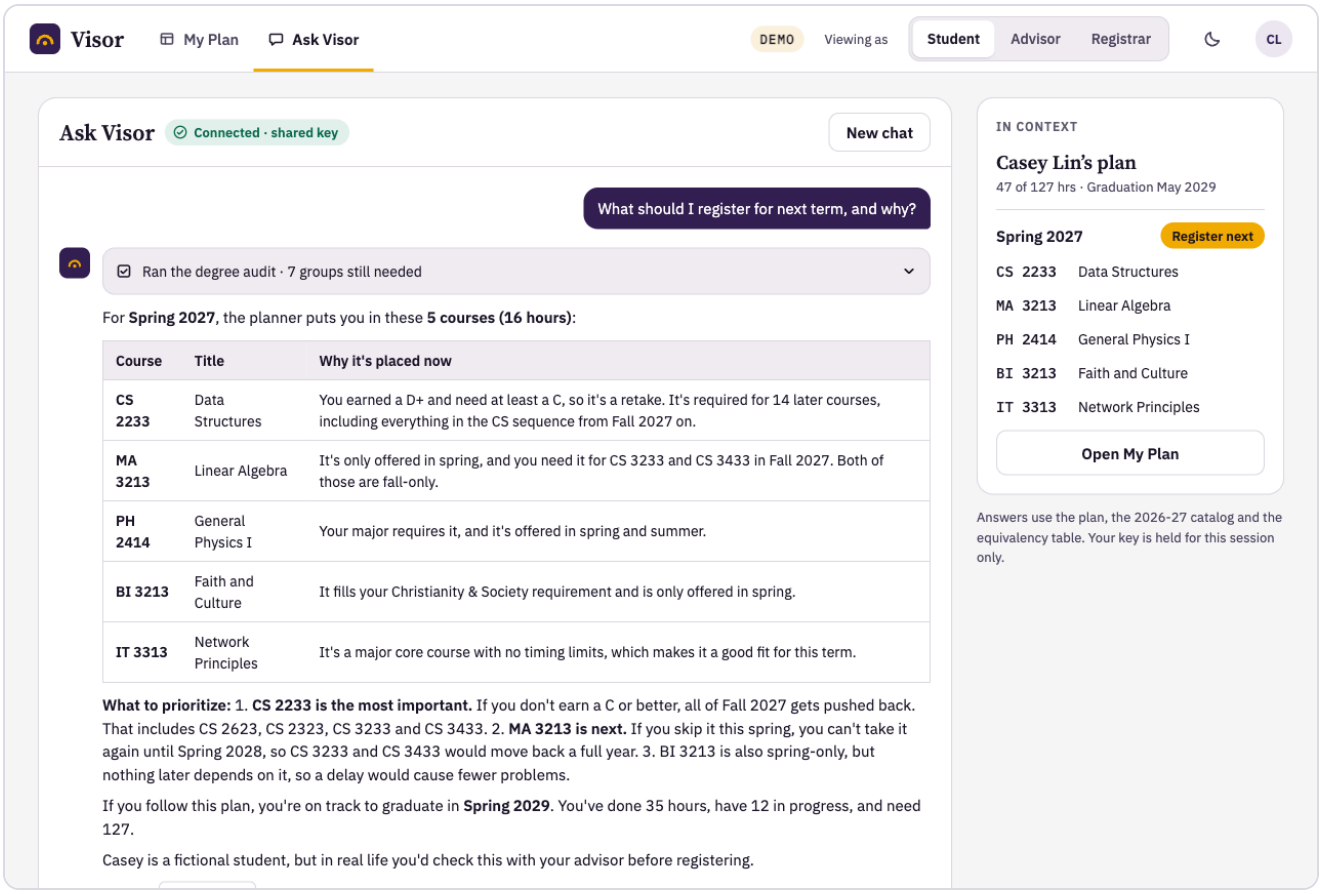
UC-C2 “What should I register for next term, and why?”

Note

Actual

A tool card (“Ran the degree audit · 7 groups still needed”), then a table of the five Spring 2027 courses with reasons that match the planner: CS 2233 retake (D+ → needs C, unlocks 14), MA 3213 spring-only, and so on. It also gave priorities, the graduation term and an advisor reminder.

Note (F5): it said skipping MA 3213 would delay *CS 3433* a year. In fact CS 3433 accepts MA 2314, which Casey already passed. It made the claim without calling the course lookup tool. The CS 3233 half of the claim is correct.

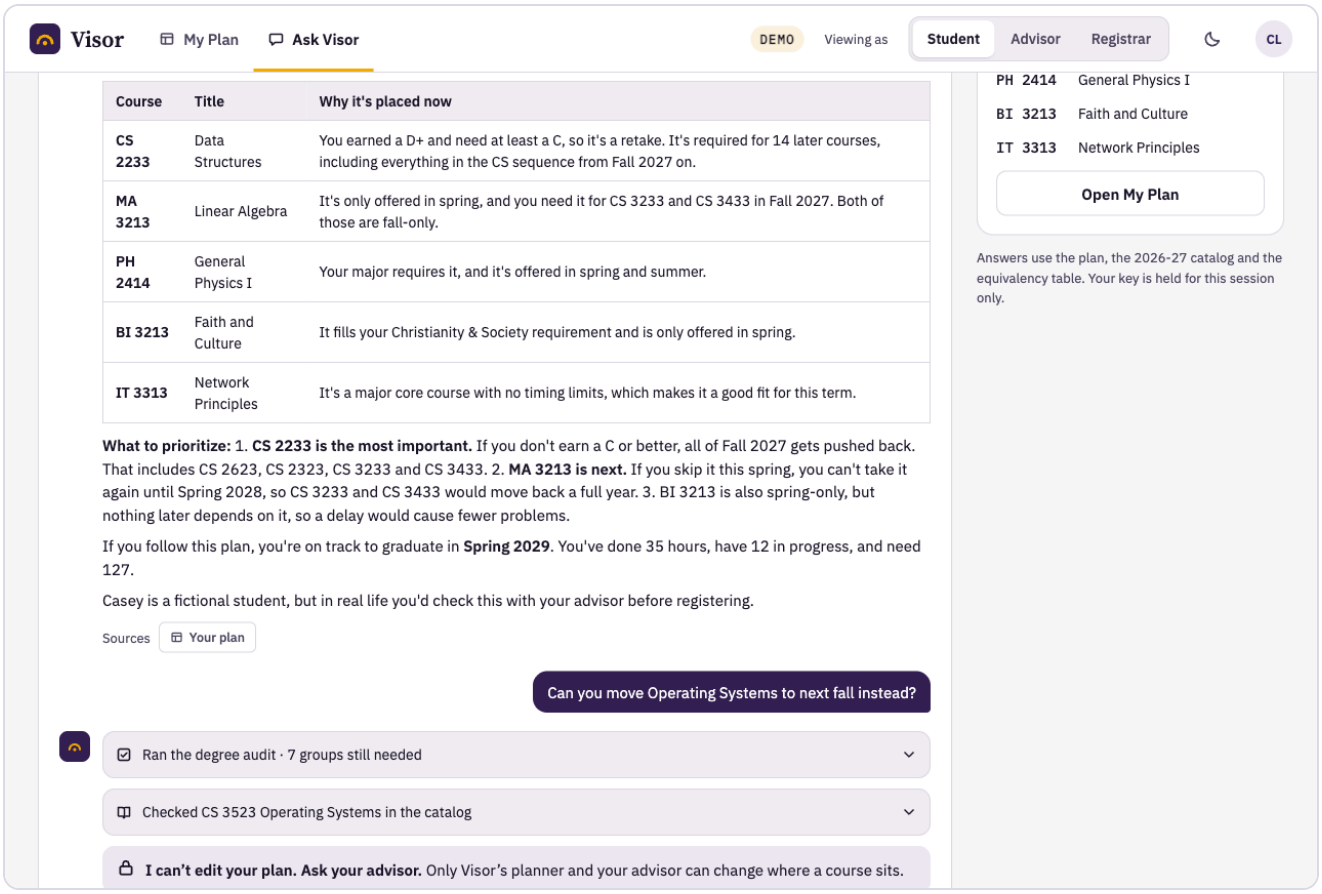


UC-C3 “Can you move Operating Systems to next fall instead?”

Pass

Actual

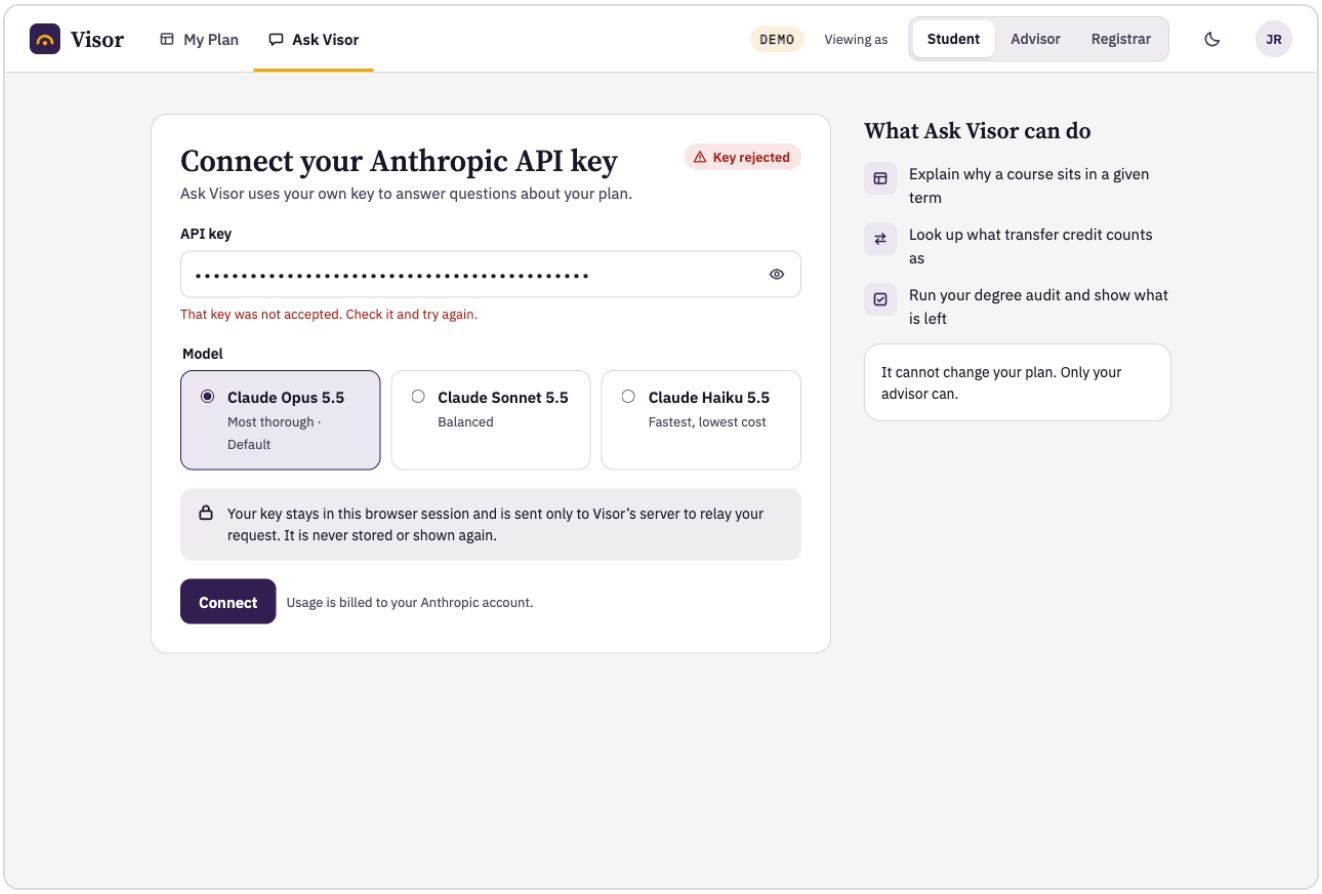
It looked up CS 3523, then replied “**I can’t edit your plan. Ask your advisor.**” and explained why fall is impossible anyway: the course is spring-only, and its prerequisite CS 2323 is fall-only and needs the CS 2233 retake. All three facts checked against `prereqs.csv` and `course_terms.csv`.



UC-C4 Student pastes a bad key

Pass

Actual “Key rejected · That key was not accepted. Check it and try again.” The key is checked with a model lookup, so a bad key costs no tokens, and the field stays masked.

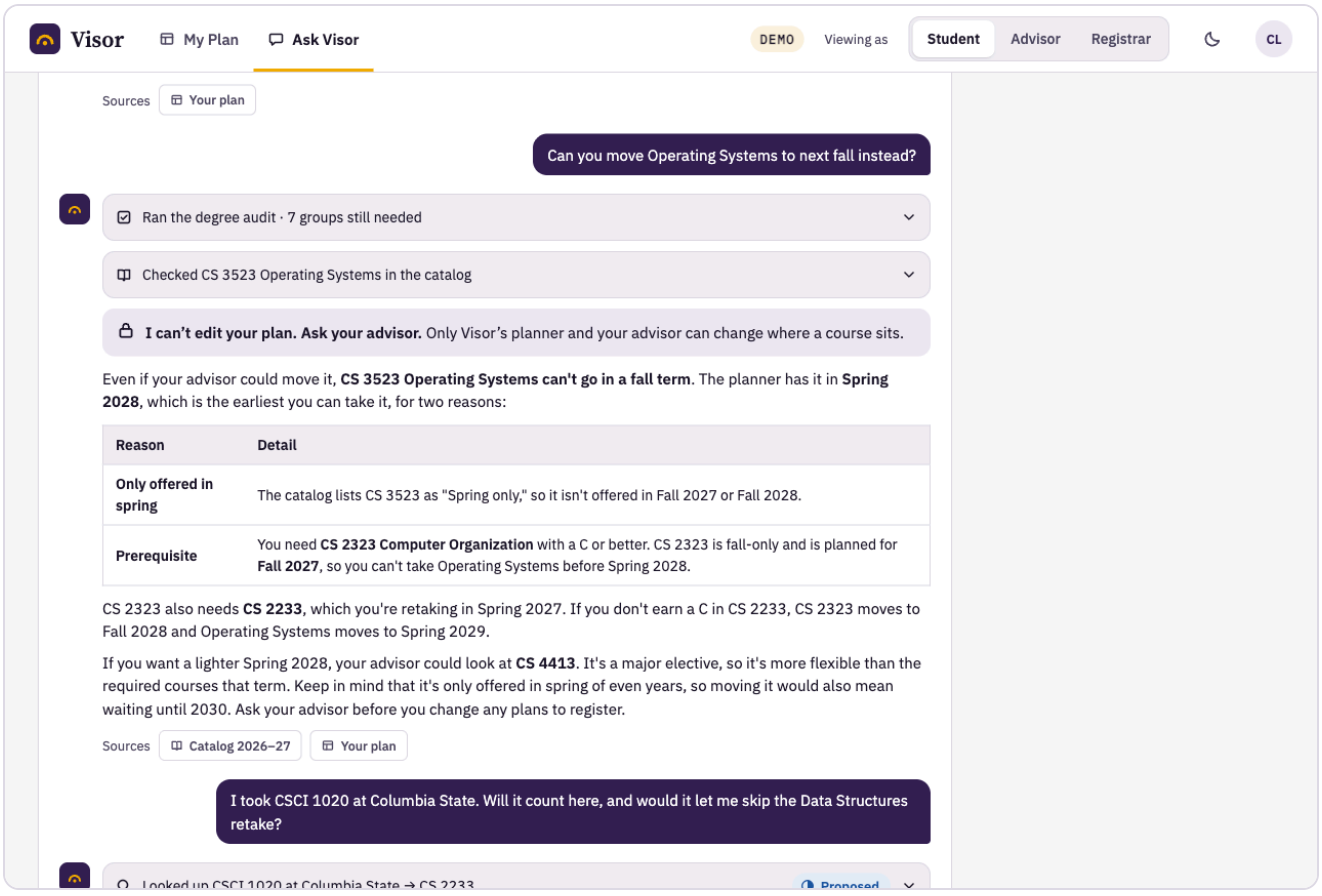


UC-C5 “Will CSCI 1020 from Columbia State count, and skip my retake?”

Pass

Actual

Tool card “Looked up CSCI 1020 at Columbia State → CS 2233 · Proposed”. The answer: **not yet**, because a proposed match doesn’t count, so keep the retake in Spring 2027. It explained what happens if the match is confirmed or denied. It also noticed the titles differ (OOP vs Data Structures), said someone will likely compare course content, and listed three questions for the advisor.



Planner edge cases (API)

These run against `POST /api/audit` directly. Every response includes `checks` from the independent checker; “0 violations” means it re-verified prerequisites, offerings, load, duplicates and class standing.

UC-E1 Same record twice gives the same plan

Pass

Actual Jordan’s record sent twice; the two JSON responses are byte-for-byte identical.

UC-E2 Summer terms turned on

Note

Actual Riley, summer off: 7 terms, graduation 2030SP. Summer on: 9 terms (adds 2027SU at 13 hrs and 2028SU at 3 hrs), graduation still **2030SP**, with a 3-hour final term. 0 violations either way.
Note (F6): turning on summer added two terms without moving graduation earlier, and it left a 3-hour summer and a 3-hour final term.

Setting	Terms	Graduation	Smallest term	Violations
Summer off	7	2030SP	15 hrs	0
Summer on	9	2030SP	3 hrs	0

UC-E3 Lower credit cap stretches the plan safely

Pass

Jordan, max hours	Terms	Graduation	Heaviest term	Violations
18 (default)	4	2028FA	18	0
12	5	2029SP	12	0
9	7	2030SP	9	0

UC-E4 Missing prerequisite gets added on its own

Pass

Actual The program requires EN 3143 but never lists its prerequisite. For Riley and Sam the planner adds EN 1223 in Spring 2027 with the reason “Needed before EN 3143”.

UC-E5 Bad input to the API

Pass

Input	Response	
Unknown program	404 “program_id must be one of cs-bs-2026-27”	OK
Term `2027XX`	400 “term code must look like 2027SP”	OK
Grade `Q`	200, treated as a failing grade and planned as a retake “earned Q”	F8
Empty body	404 (program message) rather than 400	F8
Unknown course `ZZ 9999`	Listed in `unknown_codes`, but no warning and `needs_review` stays false	F8

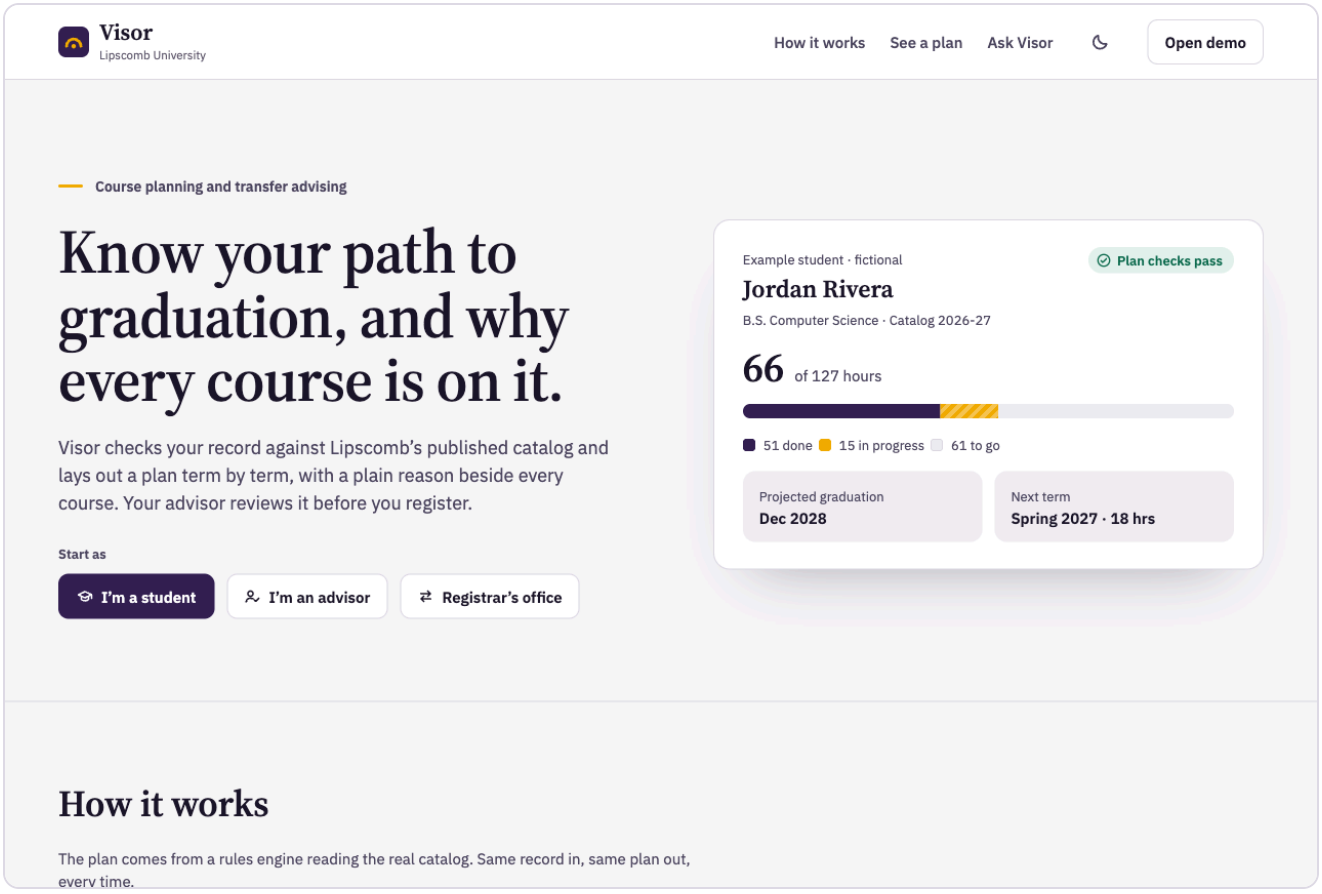
General cases

UC-G1 Visitor understands Visor from the landing page

Pass

Actual

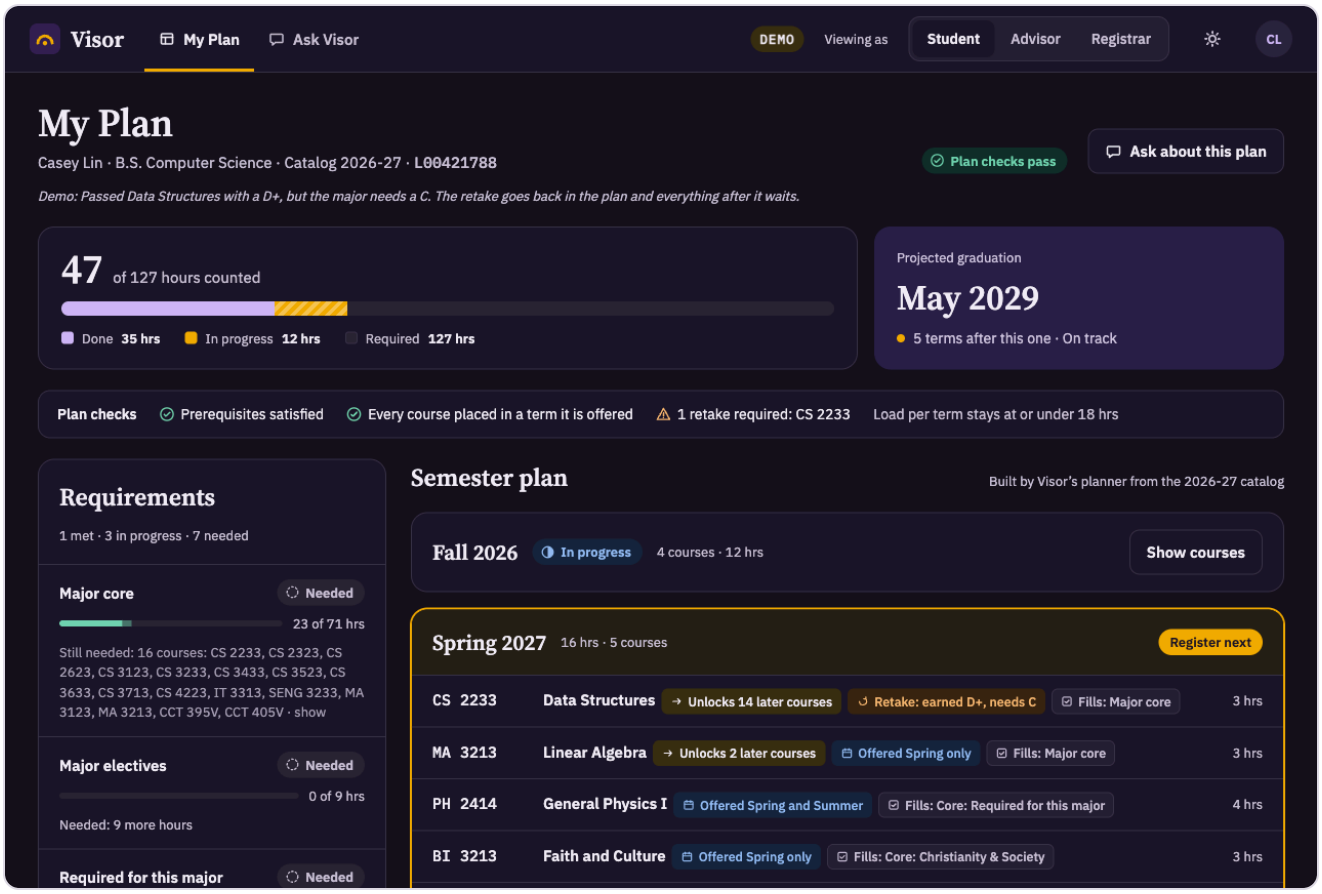
Headline, three-step “How it works”, and a live example (Jordan’s real numbers from the API, plus a live CISP 1020 lookup). Role buttons go straight to each view. The page promises “Advisors approve every plan” and “Your advisor reviews it before you register”, but the app has no approve step yet. See F1.



UC-G2 Dark mode

Pass

Actual — The toggle switches every surface; the choice is remembered. Text and status colors stay readable.

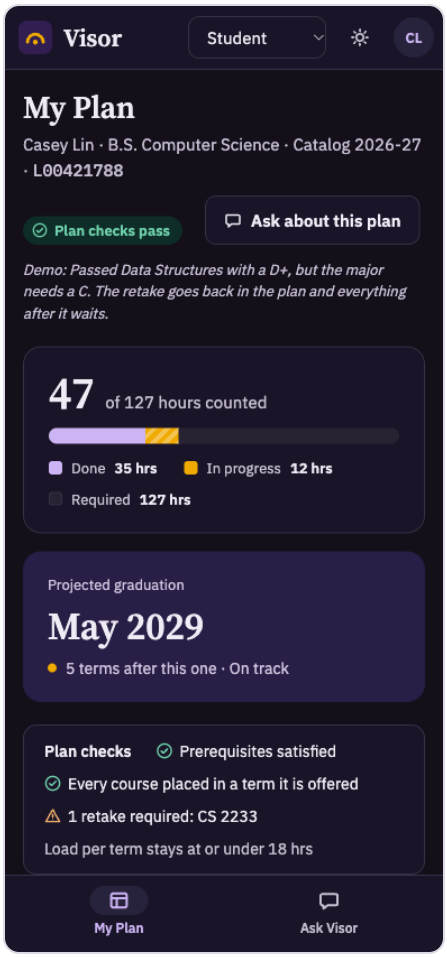


UC-G3 Phone layout

Pass

Actual

At 390 px: no sideways scrolling, a bottom nav replaces the top tabs, and the role picker becomes a dropdown. Small (F11): the “Load per term stays at or under 18 hrs” line has no icon. It is fixed text, not an actual check.



Findings and recommendations

Ranked by how much each one matters to the people using Visor. Nothing was changed during testing; these are for the team to pick from.

High F1 · Advisors can view plans but not change them

The original goal was an advisor who can *view and modify* a student's plan. Today an advisor can open any plan, but there is no way to turn on summer, lower the load, move or pin a course, or approve the plan, although the landing page says advisors approve every plan. **Next step:** a short spec (06) for advisor overrides: plan settings, "pin this course to this term", and an Approve button. Store the overrides, re-run the planner, and let `verify()` re-check the result so an override can never produce an invalid plan.

Medium F2 · The plan page hides transfer credit that doesn't count

Pending, not-transferring and unknown courses are returned by the API but not shown. Add a "Transfer credit" card to the plan listing every course with its status pill, linking unknown ones to the Transfer tab.

Medium F3 · A graduation date is shown when courses couldn't be placed

When there are unplaceable warnings, replace the date with "Can't project yet: 5 courses not placed" and keep the date out of the advisee list.

Medium F4 · The same course can be flagged twice

Each click creates another pending review. Disable the button after a successful flag, and have the flag endpoint return the existing open review instead of adding one (the 409 path already exists for confirmed rows).

Medium F5 · Ask Visor stated a prerequisite without checking

The claim that CS 3433 depended on MA 3213 was wrong, because Casey's MA 2314 already satisfies it. Either include each planned course's prerequisites in the audit tool's result, or add a line to the system prompt: call `course_info` before stating that one course depends on another.

Low F6 · Turning on summer can add terms without helping

Only open a summer term when it moves graduation earlier, and run the same last-term compaction with summer on.

Low F7 · Course code parsing on the Transfer tab

Require a letter subject followed by a number starting with a digit (`^([A-Z]{2,5})\s*-?\s*(\d[\w]*)$`), accept a hyphen, and clear the old result when showing an input error.

Low F8 · The audit API is too forgiving

Reject unknown grades with 400, return 400 for a missing `program_id`, and add a warning (and `needs_review`) for course codes not in the catalog.

Low **F9 · Free-elective hours aren't placed**

When the last term is light, say "add 4 hours of free electives here" on that term instead of only in the footnote.

Low **F10 · "Needed" on half-finished groups**

Show a third state such as "Partly done" when some hours count but the group isn't met.

Low **F11 · Load line isn't a real check**

Compare `checks.max_load` with `checks.load_limit` and show a check or alert like the other lines.

Low **F12 · No demo student shows the "needs review" path**

Add a sixth fictional student whose plan produces a warning, so the advisor demo and the Needs review chip show something.

Low **F13 · One demo equivalency is unlikely**

CSCI 1020 (Object-Oriented Programming) is proposed as CS 2233 (Data Structures). That is fine as illustration, but a closer match makes a better demo.

What held up well. Every plan we produced (five students, summer on and off, three credit caps) passed the independent checker with zero violations. Reasons on every course matched the catalog. Ask Visor refused to edit, used the planner's own data, and named what it didn't know.

Appendix: where each case is specified and tested

Spec stories are in `specs/05-design-forecast.md` (the transfer lookup is the team's `specs/03`).
Test names are in `backend/tests/`.

Case	Spec	Automated test
S1	US-5.1.1, 5.1.4	`test_every_requirement_placed_or_warned, test_start_term_and_summer`
S3	US-5.2.1	`test_every_planned_course_has_a_reason`
S4	US-5.1.5	`test_nothing_left`
A2	US-5.4.1	`test_unplaceable_course_warns`
A3	US-5.3.1	`test_audit_api_transfer_statuses`
R1–R4	specs/03	team's equivalency tests (unchanged)
C1	US-5.5.1	`test_no_key_is_refused_before_any_call`
C2, C5	US-5.5.2	`test_audit_tool_returns_the_planners_plan, test_transfer_question_uses_the_equivalency_table`
C3	US-5.5.3	system prompt; no write tool exists (verified live here)
C4	US-5.5.4	`test_errors_map_to_the_designed_states`
E1	US-5.1.3	`test_forecast_is_deterministic`
E2	US-5.1.4	`test_start_term_and_summer`
E3	US-5.1.2	`test_plan_respects_prereqs_offerings_and_load, test_verify_catches_broken_plans`
E4	US-5.2.3	`test_prerequisite_gap_is_filled`

Rebuild this PDF: `docs/user-cases/build_pdf.sh`. Screenshots are in `docs/user-cases/screenshots/`. All student data shown is fictional.